INSTRUCTIONS

GEK- 8282

IC4965

CONTROL PANEL
FOR
SCR DC DYNAMOMETER

GENERAL ELECTRIC
REQUISITION NUMBER

INDUSTRY CONTROL DEPARTMENT
GENERAL ELECTRIC
SALEM, VA.

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DESCRIPTION

IC4965 SCR D-C DYNAMOMETER CONTROL

GENERAL

The d-c dynamometer system is an adjustable voltage system using the d-c dynamometer as a conventional d-c shunt motor.

A motor-generator set consisting of a squirrel cage induction motor or a synchronous motor and a d-c generator are used in conjunction with the dynamometer as shown in Figure 1.

When the dynamometer is motoring, power is taken from the a-c line and supplied to the dynamometer through the motor-generator set. When the dynamometer is absorbing, power is returned to the a-c line.

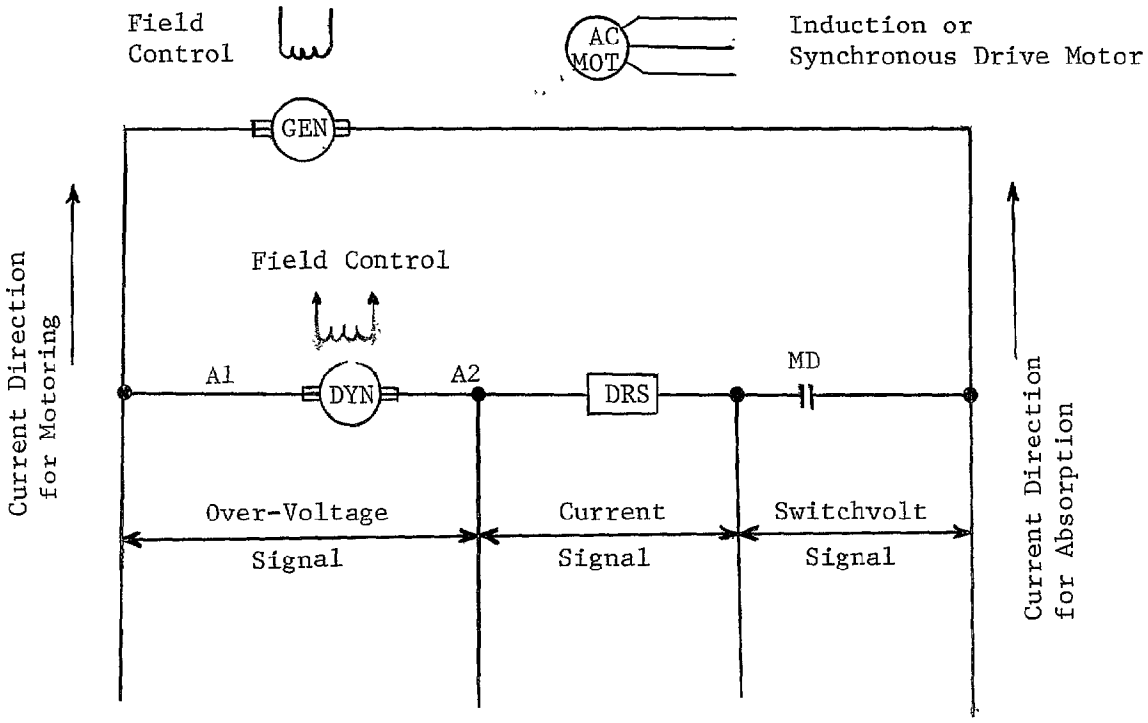


Figure 1

Control of the dynamometer and generator is accomplished by field control of both machines, using an SCR exciter for each. (See below for field adjustment.) A regulated power supply is used to obtain reference signals. Speed may be accurately controlled by use of suitable input circuits. A block diagram is shown in Figure 2.

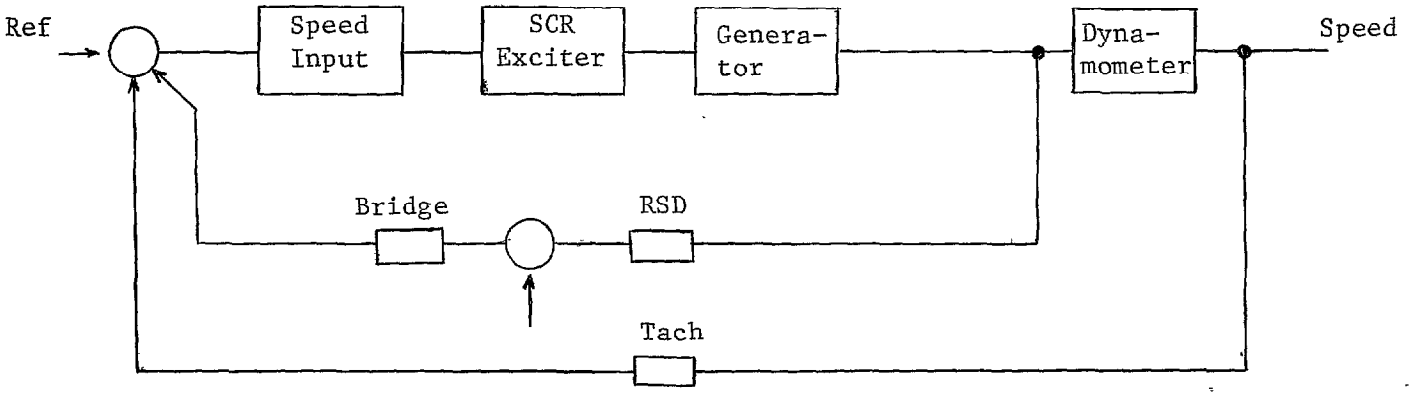


Figure 2

ADJUSTMENT

The proper procedure for adjusting the power supply, current limit circuits, stability circuits, maximum speed, dynamometer minimum field, dynamometer field current feedback, and overspeed relay are described later.

POWER SUPPLY

A regulated power supply is furnished in order to supply constant reference signals - minimizing the drift of the dynamometer operation. See power supply specifications for details. Specification numbers are shown in elementary.

SPEED CONTROL

The system is capable of closely maintaining speed with 100 percent change in load torque. A d-c tachometer-generator, coupled to the dynamometer, supplies a voltage directly proportional to speed. This voltage is compared with a voltage reference set by a speed adjustment potentiometer on the operator's panel which is connected to the regulated reference supply as shown in Figure 3.

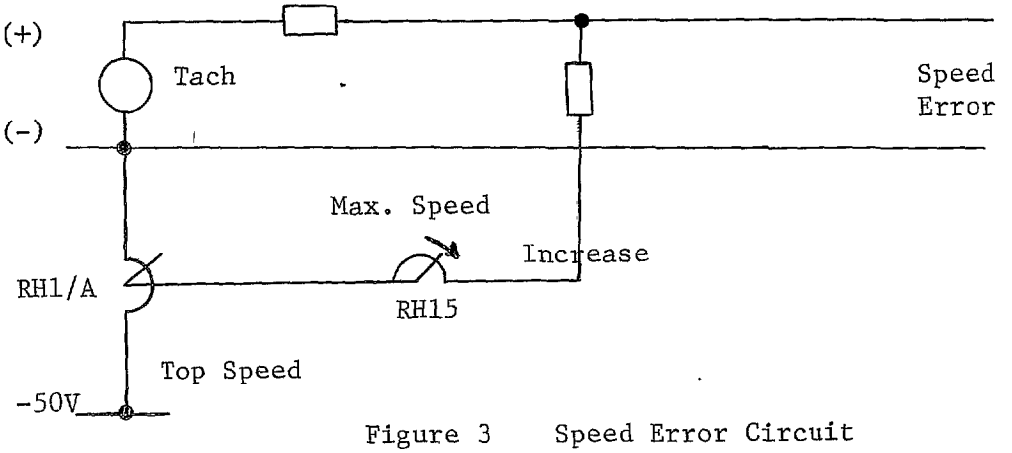


Figure 3 Speed Error Circuit

If the speed signal does not agree with the reference, the resultant error voltage will be applied to the generator SCR to correct the speed.

The SCR, which is described in the SCR manual, excites the generator field. The generator changes the speed of the dynamometer (by changing its terminal voltage) in a direction to rebalance the tachometer-generator and reference signals.

DYNAMOMETER FIELD CONTROL

The dynamometer field current is supplied by the dynamometer silicon controlled rectifier package SCRD. This SCR unit is controlled from a pre-amplifier which serves as the SCRD dynamometer field current regulator. The desired field current is set by the speed rheostat. The voltage drop across the resistor RSF in series with the dynamometer field is compared with this voltage reference. The error signal is amplified by the preamplifier and SCR in order to maintain the desired dynamometer field current.

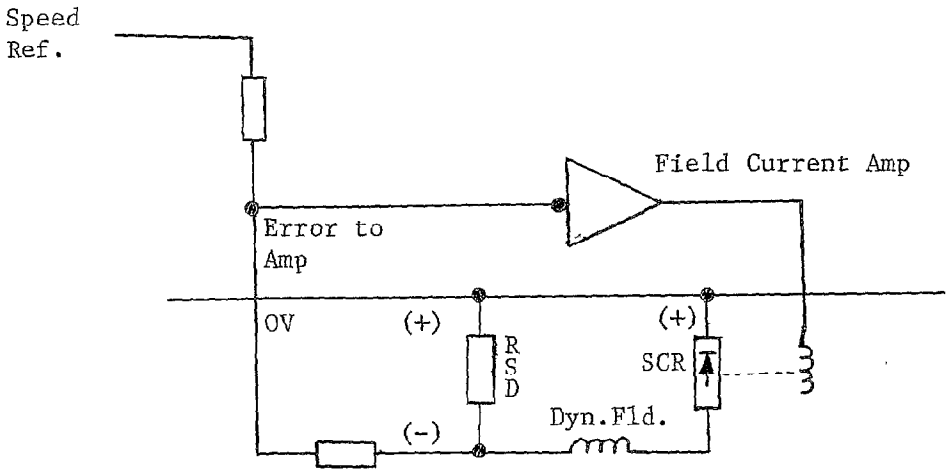


Figure 4

When a speed higher than base speed is desired, some method of changing the reference to the dynamometer field current regulator is needed. This tapering of the field is accomplished by a tapered plate on the speed rheostat. (SEE FIGURE 5)

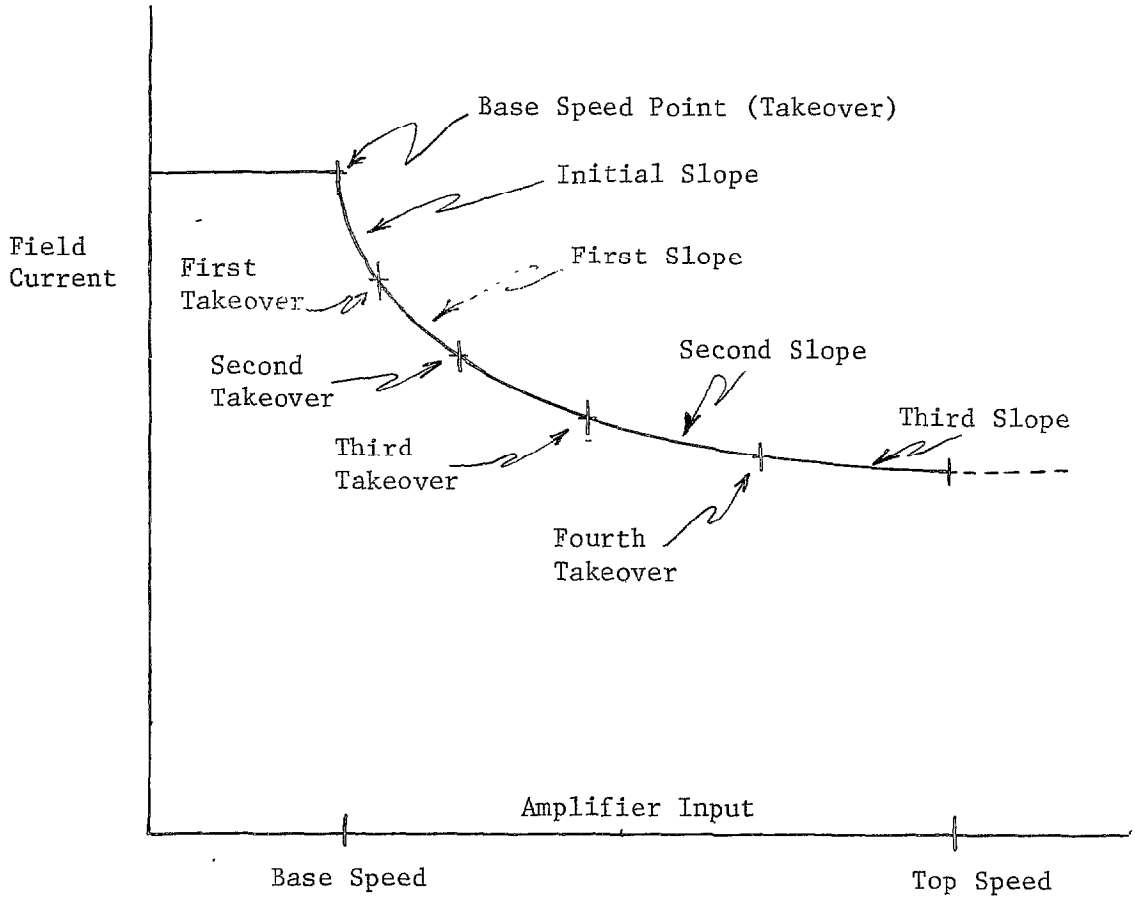
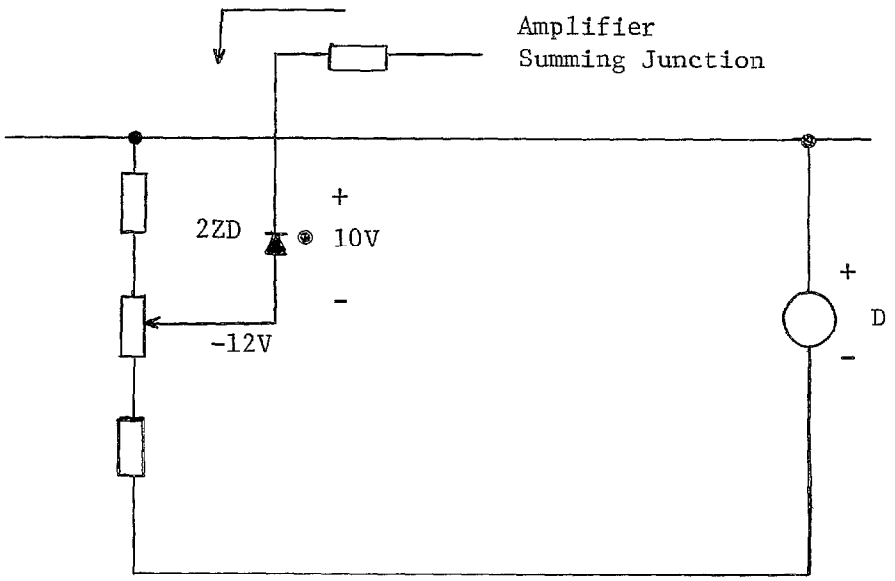


Figure 5 - Dynamometer Field Current Characteristics

If the voltage across the dynamometer armature should become too high (for instance, if the test engine should drive the dynamometer above base speed with full dynamometer field current), then the voltage limit circuit will operate through Zener diode 2ZD, causing the dynamometer field current to be lowered, as seen in Figure 6.



Current flowing out of the summing junction decreases dynamometer excitation.

Figure 6

For operation of the system above base speed of the dynamometer, in the speed regulation mode, the generator voltage is maintained approximately constant and the dynamometer field current is reduced. This is a constant horsepower type of operation, in contrast to the constant torque operation below base speed. As the dynamometer field current is lowered by the speed control potentiometer on the operator's panel when operating above base speed, the dynamometer tachometer-generator is still supplying a signal which is compared to the setting of the operator's speed control potentiometer. This error operates as before to adjust the generator field, hence dynamometer armature voltage. Therefore, regulation of speed is still accomplished by means of dynamometer armature voltage. Above base speed, the armature voltage is primarily a trimming action, in that main control is obtained by dynamometer field weakening.

CURRENT LIMIT CIRCUIT

Maintaining speed within close limits requires a high gain system. This means that a small error between the tachometer generator voltage and the reference voltage will cause a large change in motor armature voltage. This large change could cause the armature current to become excessive. Therefore, a circuit which limits motor armature current is included to protect the machinery.

This is accomplished by modifying the speed error with a signal from the armature dropping resistor (see Figure 7).

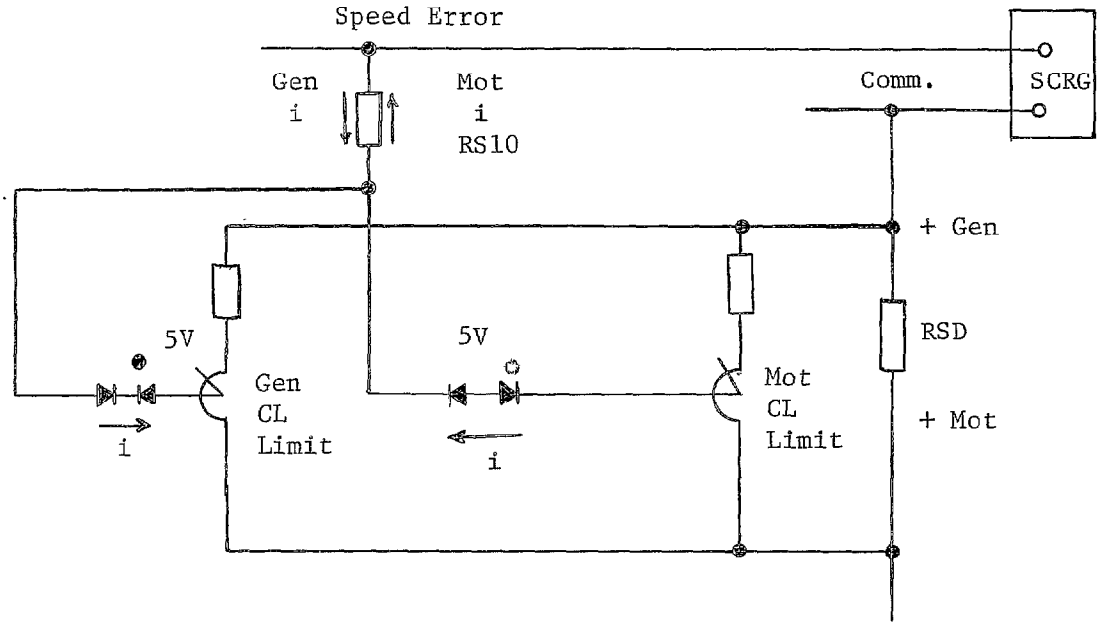


Figure 7

When the voltage of RSD causes the setting of the current limit pots to exceed 5 volts (Zener diode voltage is 5 volts), current will flow in the directions shown by the arrows. This will increase or decrease generator voltage as needed to lower the armature current. The range of current limit is 66% to 175% of full load current.

EMERGENCY STOP

If the EMERGENCY STOP pushbutton is pushed, the control will automatically be transferred zero speed reference. This will cause the regulator to operate from a speed signal modified only by current limit. The dynamometer will thereby be stopped rapidly by regenerative braking. User's protective devices, such as bearing temperature relays, cooling water pressure switches, etc., can be connected in to give an emergency stop. An ignition relay will also be dropped out, and will interrupt ignition for the test engine. During regenerative braking, the current limit circuit functions to prevent a loop circuit overcurrent.

SWITCH - VOLT RELAY

The switch-volt relay is connected across the loop circuit contactor, and will prevent closing this contactor if the voltage across its tips exceeds approximately 60 volts.

SWITCH - VOLT REGULATION

Switch volts refer to the voltage across the loop contactor when it is de-energized. Switch-volt regulation is the term applied to the function of regulating switch volts to a low value (within 5 percent of rated voltage) when the loop contactor is de-energized whether or not the dynamometer is rotating.

When the loop circuit contactor is opened, the voltage across the contactor is fed into the current preamplifier. This signal causes the current preamplifier to maintain switch-volts at approximately zero.

SPEED ERROR DETECTING RELAY (If Supplied)

The speed error detecting relay compares the actual speed of the dynamometer as measured by the tachometer-generator with the desired speed as set by the speed control potentiometer. If this difference exceeds approximately 1.5 volts, a relay will be energized and will prevent closing of the master relay, MR. This prevents the operator from attempting to reclose the loop circuit contactor in case the indicated speed, as set by the speed control potentiometer, is not consistent with the actual dynamometer speed.

OPERATIONAL AMPLIFIERS

It should be recognized that the internal circuitry of the operational amplifiers is not adapted to field repair. Therefore, it is recommended that no repair be attempted on the amplifier. If it is determined that the amplifier is defective, remove it and plug in a different amplifier. Return the defective amplifier to the factory for repairs.

SILICON CONTROLLED RECTIFIER EXCITER

This unit has been tested in the factory and has been adjusted for operation with the machines listed on the data sheet of the elementary diagram. If trouble develops or if a check is desired, follow the procedure given in the SCR manual.

STABILIZING CIRCUIT

Again, the system must have a high gain in order to obtain the accuracy desired. In high-gain systems, where a feedback is compared with a reference signal, instability may become a problem.

Instability is caused by energy storage devices and components within the closed loop system - in particular the inertia of the dynamometer and test engines, but also the armature loop circuit inductance, and generator field inductance.

Stabilization is accomplished by use of a lag-lead or "notch" circuit (see Figure 7), which is the equivalent of the simple resistor-capacitor combination shown in Figure 8.

This circuit alters the effect of input signals according to the magnitude of their rate of change with respect to time.

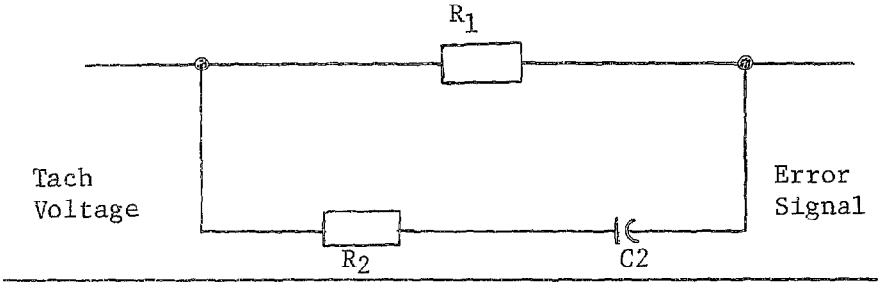


Figure 8

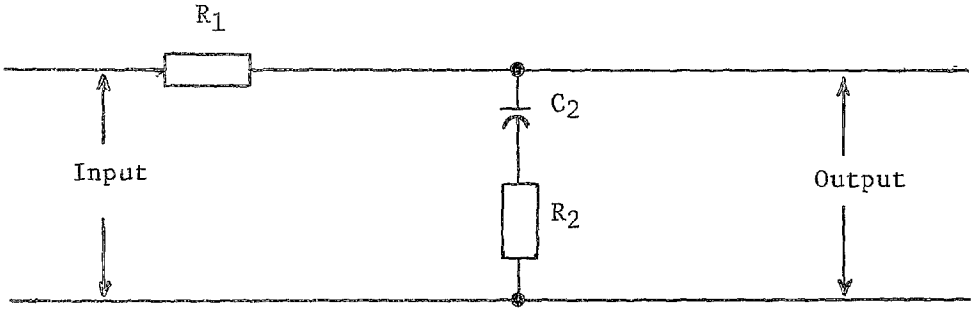


Figure 9

To adjust stability circuits, in the event of instability, two methods are provided.

1. Notch or lag-lead circuit is changed by adding more capacitance to C_2. (Figure 7) 10 MF is usually supplied in this circuit. To improve stability, add parallel capacitance. Do not add any more capacitance than necessary for satisfactory operation, or exceed 40 MF.
2. If the above method proves ineffective, then the forward gain of the system should be reduced. This is done by increasing resistance of gain pot RH16 unt desired stability is reached.

PROTECTIVE FEATURES

Included in the control system are various protective devices in addition to the EMERGENCY STOP pushbutton discussed above. Only the following items will open the loop circuit contactor:

1. Motor-generator set overspeed (centrifuge switch on motor-generator set).

2. Dynamometer overspeed (voltage sensitive relay operating from tachometer-generator). If supplied.

3. Armature loop circuit instantaneous overcurrent relay.

4. Dynamometer field loss relay.

5. STOP pushbutton.

6. Dynamometer overvoltage relay.

7. Auxiliary starters or main motor-generator set starter.

As mentioned above, the speed error detector relay and the switch-volt relay will prevent the operator from closing the line contactor unless the voltage across it is small.

OPERATIONAL TECHNIQUES APPLIED TO REGULATING SYSTEMS

The term operational imposes several conditions on an amplifier it is applied to. These are:

1. Gain is sufficiently large that input voltage and current are considered to be zero.

2. There is no internal time constant, i.e., bandwidth on frequency is very wide and rise time on step input from 5% to 95% of output is negligibly small.

The above conditions allow the following conclusions. For the configuration shown,

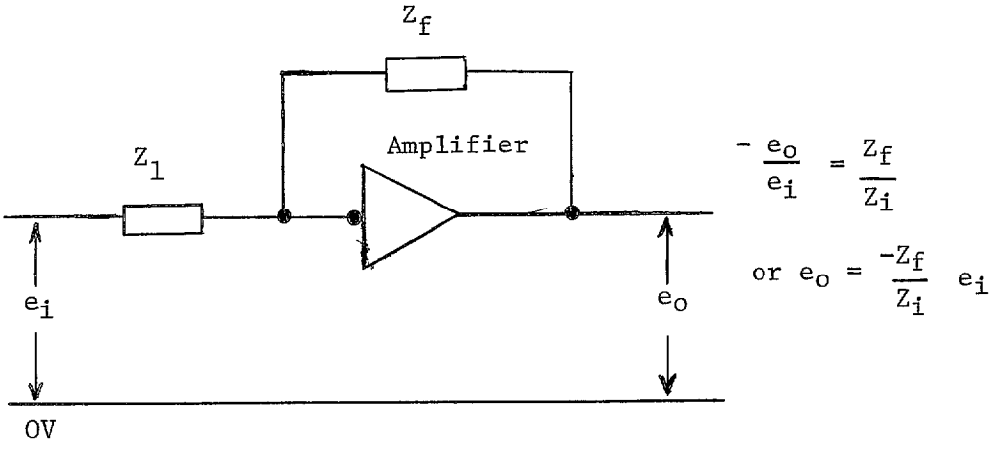


Figure 10

Z_i and Z_f can be any combination of capacitance, inductance, and resistance. The negative sign indicates 180 degree phase shift (polarity change) from input to output.

INSTALLATION & ADJUSTMENTS

Receiving and Unpacking

The control equipment is mounted in a rigid floor-mounted enclosure; and panels are crated and shipped in an upright position. When a panel is received, it should be handled from the top and kept upright.

A panel which is not to be installed immediately should be stored in a clean, dry location. When the panel is installed, the control should be unpacked carefully, making certain that no small parts are thrown away with the packing material.

Check the nameplate rating of the machine to be sure that it agrees with that given on the panel nameplate. The control is designed for use with specific equipment and must not be applied to any other without first checking with the nearest Apparatus Sales Office of the General Electric Company.

After the panel has been unpacked, it should be handled from the removable lifting angle on top provided for that purpose. To avoid undue strains on the panel, sling the lifting rope through the holes near the ends of the lifting angle.

Clean the inside of the control panel with a brush, soft cloth, or dry compressed air. All dirt, dust, or bits of packing material, which interfere with successful operation of the panel devices, should by all means be removed from the panel. You should likewise take care during the cleaning operation to prevent any dirt from being blown into the inaccessible spaces of the devices.

Installing the Panel

When the panel is to be installed, it is essential that it be securely fastened in a true, upright position on a level surface to permit proper operation of the devices mounted on the panel.

The closing case is mounted on two floor sills, one each across the back and front of the enclosing case. Floor sills consist of channel irons which are four inches deep and one and three-quarters inches high. The four-inch dimension is horizontal. On single-case installations the floor sills may be removed and the enclosing case bolted to the floor. On lineups of more than one case it is recommended that floor sills be used to give rigidity to the panel. Mounting holes for one-half-inch bolts are provided in each sill. The outline for the particular panel to be installed should be checked to determine the location of the mounting holes. It is recommended that the one-half-inch mounting bolts be grouted in the floor in the proper location for fastening the panel.

There are several possible methods of installing the floor sills. The most frequently used methods are described below:

If the panel will be installed on an existing floor, grout the one-half-inch anchor bolts in the floor and slip the panel over them. Check level position of the sills with a machinist's level, and shim the sills if necessary. Check that the surfaces of the front and back sills are in a plane, so the angle iron supports of the panel are plumb. If the panel is installed in this manner, there will be a two-inch gap between the floor and sides of the panel enclosure. This gap should be closed to prevent accumulation of dirt under the panel, which in due time may interfere with the proper operation of the panel devices.

If the concrete floor will be poured at the time that the panel is installed, grout the one-half-inch anchor bolts in the sub-floor, and slip the panel over them. Check alignment of the floor sills with a level and make sure that the panel supports are plumb. Then pour concrete around the floor sills. This should be done carefully, so as not to splash concrete on the panel devices. Care should also be taken to finish the surface of the concrete so there will be no interference of the enclosing case door; and so there will be a tight joint between the floor and the enclosure side, thus preventing the entrance of dirt at the bottom of the panel.

If the concrete floor is to be poured before the panel will be installed, a dummy set of floor sills can be used. These dummy sills can be made of wood of the proper size and length. The dummies should be set in place, together with the one-half-inch anchor bolts, before the concrete is poured, and then replaced with the actual sills, after the panel is received.

Provisions should be made for incoming and outgoing cable and control wires before the panel is in place. In all cases the outline for the panel being installed should be used to determine the space available, and the panel interconnection diagram should be checked for the number of wires and cables, which will be installed. The cables normally enter the bottom of the panels.

Preparing the Panel for Operation

Before you can operate the panel even for a tryout without power, all devices must be placed in fully operative condition. The armatures of some of the relays and contactors may have been tied or blocked to prevent damage during shipment. Remove all of these ties and blocks before operating devices.

Sealing surfaces of unplated and laminated a-c contactor and relay magnets, as well as the ends of shafts, are coated with a rust preventative. This protection against rust is a heavy grease which does not interfere with the proper operation of the devices, but which encourages the accumulation of dust and dirt which may ultimately prevent proper device operation.

Therefore, before operating the panel, you should wipe off all rust preventative. Magnet sealing surfaces may then be coated with a thin film of light machine oil.

Operate each device and relay by hand to see that the moving parts operate freely without binding. Make sure that all electrical interlocks are clean and make good contact when closed. Relays and contactors are carefully adjusted and tested at the factory. If adjustment information for individual devices is needed, however, it may be found in individual instructions for each device.

With the power leads disconnected at the machine terminals, but with all devices properly readied for operation, apply power and operate the panel to make certain that its operation is in agreement with the elementary diagram.

The leads may then be connected and the equipment put into operation.

Field Taper & Field Current Regulator

The field current regulator and taper has been factory set to calculated values of field current. Some slight field adjustment may be necessary. Dynamometer terminal voltage should stay between 100% plus 10% minus 0% (example: for 250 volt machine the voltage should remain between 250 and 275 volts) at any speed after transients have decayed. See the data sheet of the elementary for the end points on the dynamometer taper curve.

The following checks should be made when the dynamometer is not running. When the speed potentiometer is turned very fast, the field current will probably appear as shown. (Figure 11).

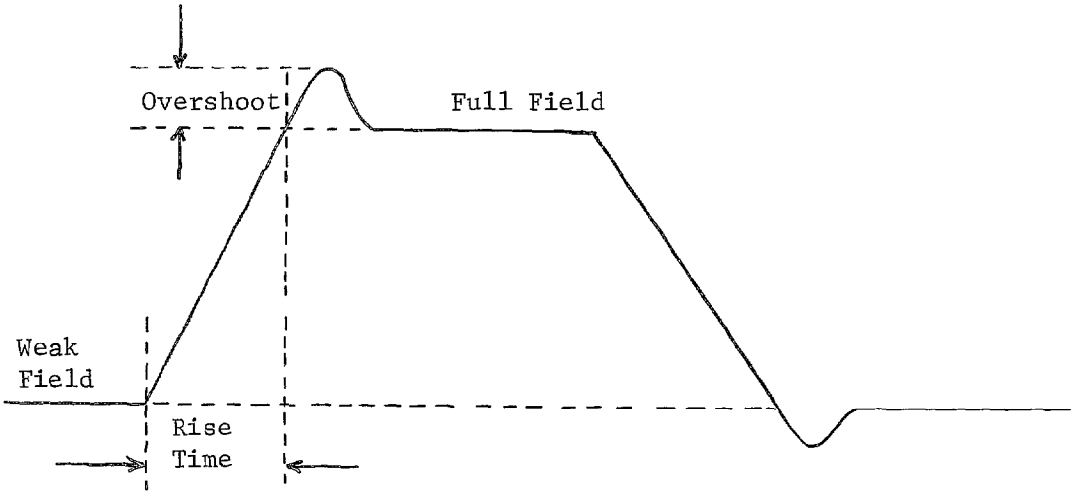


Figure 11

Increasing the ohms in RH104 will reduce the overshoot. Correct adjustment is obtained when the overshoot is less than 5%.

It may be necessary later to adjust RH105 to give faster or slower response for small steps.

The field current response should not be able to change faster than the maximum rate of change of speed.

If a machine requires 5 seconds to accelerate from base speed to top speed, then field current should change from full field to weak field in 5 seconds or longer. Depending on the generator field hot drops, some difficulty may be experienced in holding current limit on very fast changes in speed reference potentiometer setting above base speed. On acceleration voltage may be high. (See Figure 12).

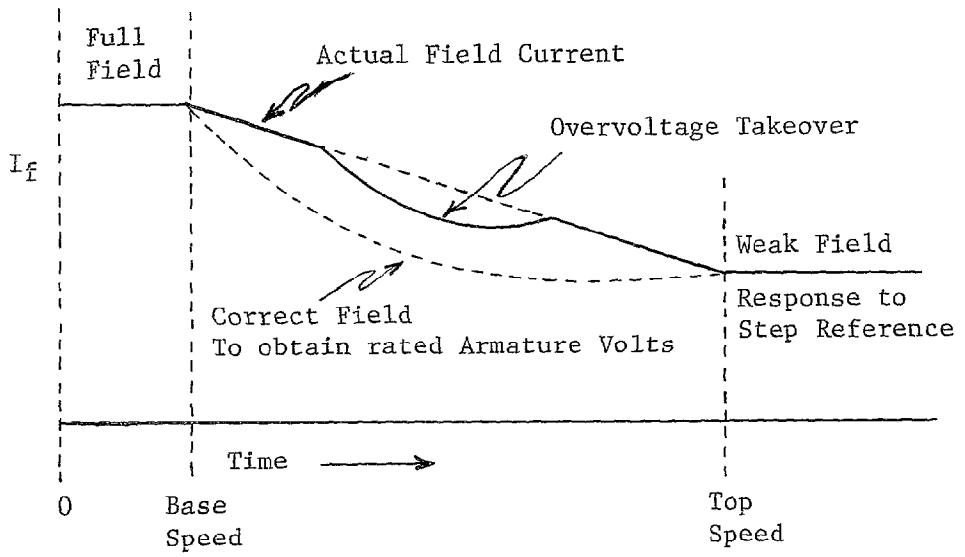


Figure 12

On deceleration voltage may also be high.

If the overcurrent relay trips, then the field may have to be slowed even further. Overvoltage takeover will then control field on rapid changes in reference.

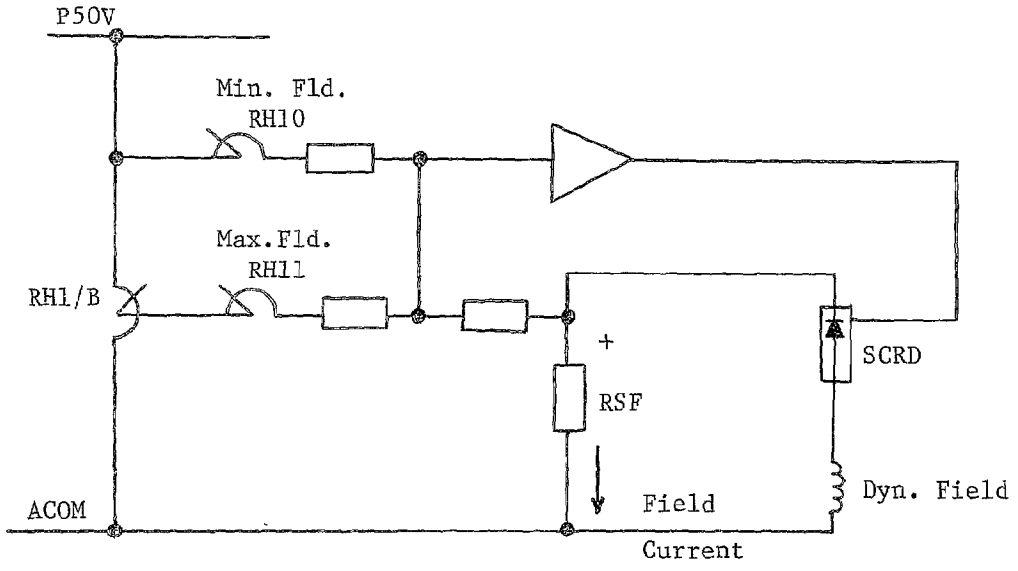


Figure 13

This tapering of the field is accomplished by a tapered plate on the speed rheostat.

Adjust max. and min. field as follows:

RH10 must be adjusted first with RH1 fully CW.
From the equation below, calculate the approximate setting for RH10.

WF = Top Speed Field Amp
FF = Base Speed Field Amp

$$RH10 = \frac{33K \times 50V}{WF \times FRS} - 220K$$

From the following equation, calculate the approximate setting for RH11.
When adjusting RH11, RH1 must be fully CCW.

$$RH11 = \frac{33K \times 50V}{(FF-WF)\ FRS} - 100K$$

A sample calculation for RH10, RH11 from the given values:

FF = 10 Amperes
WF = 2.3 Amperes

$$RH10 = \frac{33K \times 50V}{2.3} - 220K = 500K$$

$$RH11 = \frac{33K \times 50V}{7.7} - 100K = 114K$$

Overvoltage Takeover

1. With the main contactor energized and the drive running, adjust RH1 for base speed.

2. With the zero bias pot (see speed regulator portion of elementary) increased speed slowly. Monitor dynamometer voltage. When it reaches takeover voltage (approx. 120% generator voltage), adjust overvoltage pot (see dynamometer field regulator) until dynamometer field begins to weaken.

To set maximum speed control:

1. Set RH15 full CCW (Resistance in). Rotate RH1 fully CCW.

2. Start system, dynamometer motoring, and set percent rotation of RH1 from curve on sheet OB for base speed.

3. Adjust RH15 until base speed is reached.

4. Advance RH1 and check for correct top speed at full CW rotation. RH15 may require trimming at this point.

Speed Regulator

Response of the speed regulator should be in accordance with specifications pertaining to this drive. In the absence of specific details, the following is standard.

On 50% impact load, speed droop should not exceed 25% of top speed and recovery to 1% of operating speed above base speed or 1/2% of base speed below base speed shall be 5 seconds or less.

Response will also determine the amount of scale fluctuation. In general, faster response results in greater scale fluctuation.

1. Turn TSW to speed.
2. Keep current limit set low.
3. Turn RH1 to zero speed.
4. Close loop.
5. Advance speed pot slowly thru the complete speed range to check stability.
6. At a speed just below base speed, put in a small speed step by rapidly twisting the speed knob. The current should <u>not</u> reach current limit.
7. Response should be as shown below. (Figure 14)

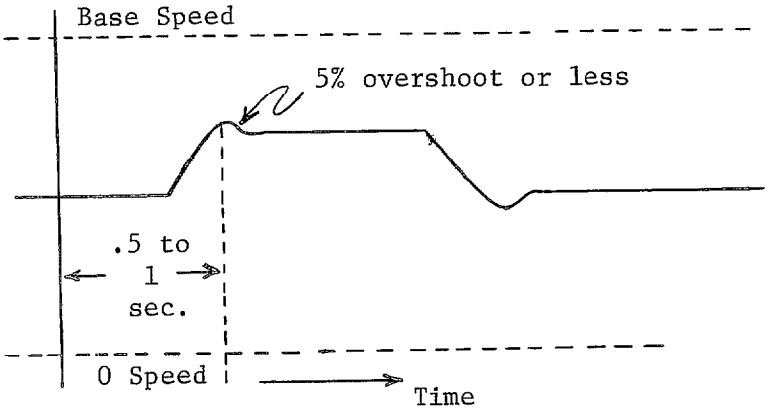


Figure 14

8. Adjust stability if necessary with gain pot.
9. Turn up current limit.

10. Apply an impact load, if possible.
Response should appear as shown below (Figure 15).

The numbers to apply to this figure are given at the beginning of this section.

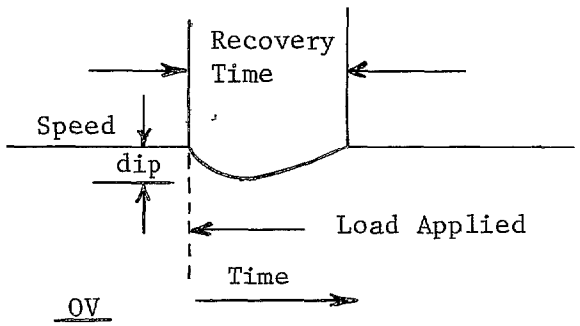
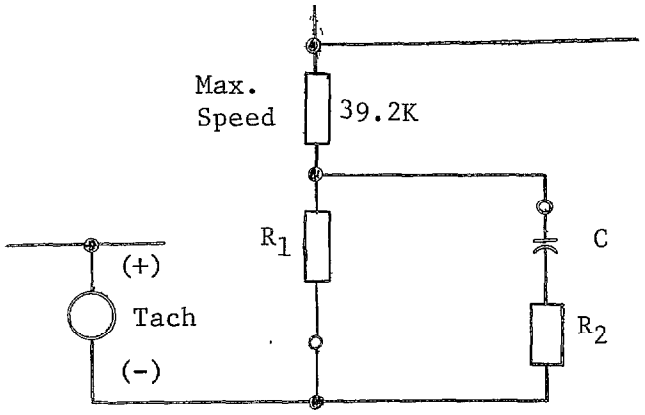


Figure 15

11. From a low value of speed, try emergency stop.

12. Advance speed and try again. Keep repeating until a stop from top speed is made successfully.

13. If reversal of rotation on stop is a problem, the tach lead circuit should be changed. (See Figure 16) This may be accomplished by changing the value of capacitance.



$(R_1 + R_2)\ C = \text{Lead}$

$R_2C = \text{Lag}$

$3 < \frac{\text{Lead}}{\text{Lag}} < 10$

Figure 16

The stopping characteristic is shown below. (See Figure 17)

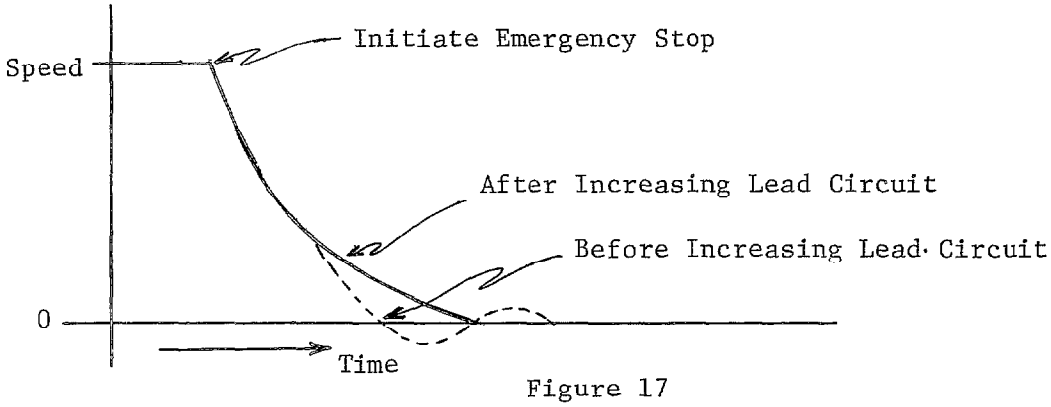


Figure 17

OPERATING INSTRUCTIONS

The sequence of operation is outlined below:

1. Turn the reversing switch to either FORWARD or REVERSE. This, of course, determines the direction of rotation of the dynamometer, and energizes the control circuit.

2. Press the motor-generator set START pushbutton. The motor-generator set will start, and will automatically sequence the dynamometer system auxiliaries to start, including the blower motors, pump motors, and SCR exciter. Each of these auxiliaries is protected with an overload relay or fuse. The motor-generator set STOP pushbutton will stop all of these auxiliaries and exciters as well as the motor-generator set.

3. Turn the transfer switch to SPEED CONTROL. It is more desirable to start up under SPEED CONTROL than under TORQUE CONTROL or CURRENT CONTROL.

4. Turn the speed control rheostat to MINIMUM.

5. Push the dynamometer START pushbutton, thus closing the armature loop circuit contactor.

6. The speed control rheostat can now be advanced to obtain the desired dynamometer speed.

7. During any changes, the operator should watch the RPM indicator, voltmeter, and ammeter in order to check operation of the system.

8. The test engine can now be energized and automatically loaded.

9. If it is desired to protect the equipment under test from too high a torque while the dynamometer is motoring, the motor current limit potentiometer may be adjusted counterclockwise. This will permit the current limit circuit to take over at lower value of current.

10. For normal stopping of the system, the load should be removed from the dynamometer and the speed reduced to zero by means of the speed control potentiometer. The dynamometer STOP pushbutton can then be pressed to open the loop contactor.

11. For emergency stopping, press the emergency stop pushbutton. This will rapidly bring the drive to rest by regenerative braking. The RESET pushbutton must then be depressed before the drive can be started.

12. Indicating lights tell the operator when the motor-generator set is running or stopped, and when the loop contactor is open or closed. The loop contactor "open" light will not come on until all protective devices are in their normal operating condition. Furthermore, some systems are supplied with additional indicating lights or annunciator drops to indicate that a protective device (described above) has operated.

MAINTENANCE

All electrical apparatus will provide maximum trouble-free service if given the benefit of preventative maintenance inspections and periodic cleanings. Although the frequency of the inspection periods will depend on operating conditions, it is important that a definite inspection routine and inspection schedule be followed. Again, maintenance instructions in the individual device instructions are most helpful.

Renewal Parts

Order renewal parts by part number and description, and identify the equipment for which they are required. Structural parts and complete operating devices are identified on the equipment drawing list. They are available on request.